

STA 2503 — Fall 2026

Mathematical Finance

Project 1 – Discrete-Time v.s. Continuous-Time Financial Models

- The following problems are due on Friday, Oct 9th, at 11:59 PM.
- Please use the latex template to submit this homework.
- Page limit: 15

Part I: Scaled Random Walks and Brownian Motion

In class, we showed that a Brownian motion can be thought of as the limit of a scaled random walk. To explore this idea:

1. Write down the definition of a Brownian motion W_t .
2. Simulate several sample paths of a Brownian motion W_t over $t \in [0, 1]$ using a discretization $N_1 = 10,000$.
3. Simulate sample paths of a symmetric random walk with steps ± 1 and probability $1/2$, scaled by $\sqrt{1/N_2}$ with $N_2 = 10$, over the same time horizon $[0, 1]$.
4. Repeat the random walk simulation with increasing number of steps (e.g., $N_2 = 50, 200, 1000$).
5. Plot the results and compare visually with the Brownian motion paths.
6. Discuss your observations: how does the scaled random walk behave as N_2 becomes large?

Part II:

In this project, you are asked to apply two different models to the stock price of NVIDIA (ticker symbol: NVDA):

- The **binomial model** (Cox–Ross–Rubinstein framework), and
- The **geometric Brownian motion (GBM)** model.

The aim is to understand how stock price dynamics can be modeled in both discrete and continuous time, and to compare their application in option pricing.

Background: The CRR Binomial Model

The Cox–Ross–Rubinstein (CRR) binomial model is a discrete-time model for stock price dynamics. It assumes that in each small time step Δt , the stock price can move either:

$$S_{t+\Delta t} = S_t \cdot u \quad (\text{up move}), \quad \text{or} \quad S_{t+\Delta t} = S_t \cdot d \quad (\text{down move}),$$

where $u > 1$ is the up factor and $d < 1$ is the down factor. In the CRR specification:

$$u = e^{\sigma\sqrt{\Delta t}}, \quad d = \frac{1}{u},$$

where σ is the volatility.

Under the risk-neutral measure, the probability of an up move is:

$$p = \frac{e^{r\Delta t} - d}{u - d},$$

where r is the risk-free interest rate.

By iterating this process N times over a time horizon T , we obtain a binomial tree of possible stock price paths. At maturity, option payoffs (e.g., for a call option, $\max(S_T - K, 0)$) are evaluated at the terminal nodes, and the option price is determined by backward induction using risk-neutral probabilities.

This discrete-time framework converges to the continuous-time Black–Scholes model as $N \rightarrow \infty$.

Tasks

1. **Data Collection:** Record the most recent daily NVIDIA stock price (e.g., from a financial website such as <https://www.investing.com/equities/nvidia-corp>). For reference, as of September 14, 2026, the stock price was approximately \$212.13.
2. **Binomial Model Setup:** Using the CRR framework, estimate the parameters u , d , and p for the binomial model.
 - (a) **Binomial Model Implementation:** Build a binomial tree to simulate possible stock price paths. Use reasonable assumptions for:
 - risk-free rate r (e.g., 4%),
 - volatility σ (e.g., 30% annualized),
 - number of steps N (e.g., 50 or 100).

- (b) **Binomial Option Pricing:** Extend the model to price a European call option with strike K equal to the current stock price. Compare the results obtained with different values of N .
- (c) **Python Algorithm for a 1-Year Call Option (Binomial):** Write a Python program that uses the binomial model to compute the price of a European call option with:
- Strike price K equal to today's Amazon stock price,
 - Maturity $T = 1$ year from today,
 - The same parameters r , σ , and N as in the previous steps.
3. **Geometric Brownian Motion (GBM) Model:** Assume that the Amazon stock price follows a geometric Brownian motion:
- $$dS_t = \mu S_t dt + \sigma S_t dW_t,$$
- where μ is the drift, σ is the volatility, and W_t is a standard Brownian motion.
- (a) Write down the solution for S_t .
- (b) Implement a simulation of stock price paths under GBM using Python (e.g., with Euler-Maruyama discretization).
- (c) Use the simulated paths to estimate the price of a 1-year European call option with strike equal to today's stock price (via Monte Carlo simulation).
4. Compare your GBM-based option price with the binomial model result. Compare your simulated result with the Black-Scholes formula and state your findings.

Deliverables

Submit a report including:

- A clear description of your assumptions,
- The construction of the binomial tree,
- Sample simulations of stock price paths under both models,
- Python code for both the binomial and GBM approaches,
- Numerical results for the 1-year European call option price,
- A short discussion comparing the binomial and GBM results.